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Cross-Model LLM Code Review: Should you use Claude to review Codex or vice versa?

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code & data

1. Motivation and research questions

Developers increasingly run two coding agents together: one writes a draft, and the other reviews it before the change lands. Leaderboards rank single models in isolation; they do not say which of two available agents should write, which should review, or whether the second pass is worth its cost and time.

We treat this as a role assignment problem between a writer and a reviewer, and measure what a single review pass changes under a common hidden-test evaluator.

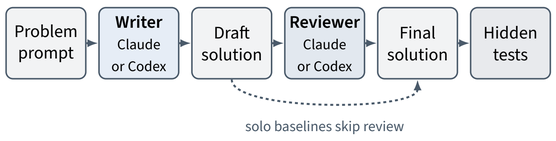
RQ1 Does a reviewer pass improve overall correctness?

RQ2 Does its value depend on which model writes and which reviews?

RQ3 When does the reviewer fix the draft, and when does it regress it?

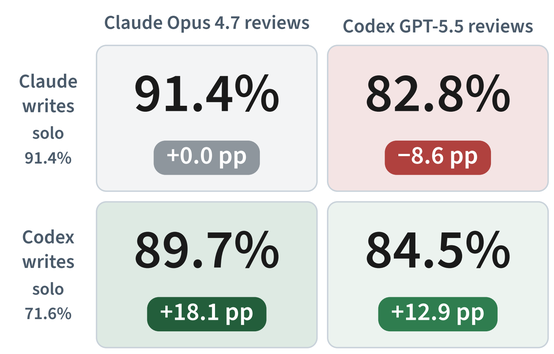
RQ4 Are correctness gains worth the added cost and latency?

2. Study design



- 116 recent hard and medium LiveCodeBench tasks (self-contained Python contest problems scored against hidden tests), all released after both models' training cutoffs; each task has a valid artifact under all six conditions (complete-case pooling).
- Claude Opus 4.7 (Claude Code CLI) and Codex GPT-5.5 (Codex CLI), both at high reasoning effort.
- Static review: the reviewer reads the problem and the writer's draft and must re-emit a complete final program; there is no approve or no-change action. It cannot run code, query a test runner, or inspect execution traces.
- The benchmark scores only the final artifact, so for a fixed writer, the difference between the solo and reviewed conditions is the review pass itself.
- Metrics: pass rate (bootstrap 95% CI), SID, reviewer fixes and regressions; exact McNemar paired tests with Benjamini-Hochberg (BH) correction, defined in the results panels.

3. Pass rate by writer and reviewer



Answer: have Claude Opus 4.7 review Codex GPT-5.5 drafts (+18.1 pp); the reverse loses 8.6 pp. No reviewer improves Claude Opus 4.7 drafts.

4. Complete results

Cond.	Pass rate [95% CI]	SID	Cost	Latency	Reg. rate
Solo baselines					
A	.914 [.862, .957]		$.226	86.2 s	
O	.716 [.629, .793]		$.190	38.5 s	
With a review pass					
AO	.828 [.759, .897]	−.086	$.382	118.0 s	.112
OA	.897 [.836, .948]	+.181	$.443	112.4 s	.043
AA	.914 [.862, .957]	+.000	$.389	135.8 s	.026
OO	.845 [.776, .905]	+.129	$.312	67.9 s	.052

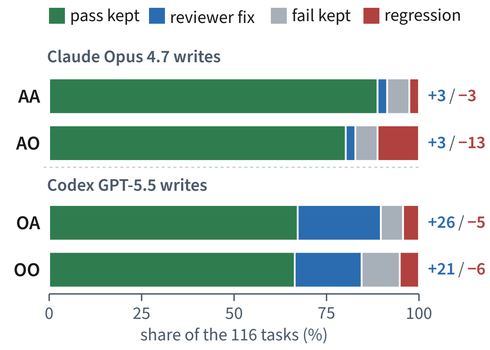
Letters are writer then reviewer (A = Claude Opus 4.7, O = Codex GPT-5.5). SID: change vs the writer's solo baseline. Reg. rate: share where the baseline passed but the reviewed run failed.

5. Six conditions

ID	Writer	Reviewer	Role type
A	Claude Opus 4.7	–	Solo baseline
O	Codex GPT-5.5	–	Solo baseline
AO	Claude Opus 4.7	Codex GPT-5.5	Cross-model
OA	Codex GPT-5.5	Claude Opus 4.7	Cross-model
AA	Claude Opus 4.7	Claude Opus 4.7	Same-model
OO	Codex GPT-5.5	Codex GPT-5.5	Same-model

Two-letter labels are in writer-then-reviewer order: AO is Claude Opus 4.7 output reviewed by Codex GPT-5.5.

6. What the reviewer changed



OA fixes 26 of the 33 failures Codex leaves and regresses 5 (net +21); AO fixes 3 of Claude's 10 and regresses 13 (net −10).

7. Statistical support

Comparison	First only	Second only	p_{BH}	
OA vs O	26	5	.0010	significant
OO vs O	21	6	.0222	significant
AO vs A	3	13	.0456	significant
AA vs A	3	3	1.000	n.s.
AA vs AO	12	2	.0323	significant
AO vs OA	5	13	.1444	n.s.

First/second only: tasks passed by only that condition (McNemar discordant pairs). AO vs OA does not survive correction; the 19.8 pp baseline gap means reviewer strength and review direction cannot be fully separated.

8. Helpful reviews edit; harmful ones rewrite

Targeted edits that keep the writer's structure

Examples: restoring deleted state in a segment tree while keeping the rest of the writer's interface; removing stray code fences so an otherwise working solution could be parsed.

Rewrites that discard a working draft

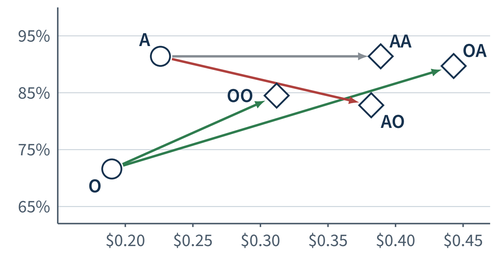
Examples: replacing a passing median-window solution with a heap rewrite that fails hidden tests; removing a range guard that protected an edge case.

Discussion and implications

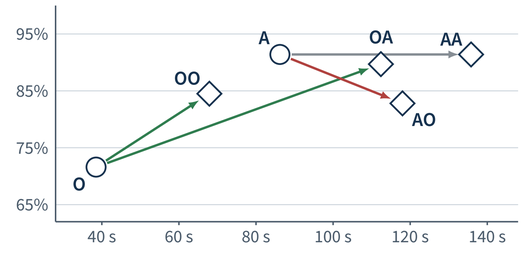
- Review pays where the writer under-verifies: Claude's long solo turns (86.2 s vs 38.5 s) suggest heavy first-pass checking (AA +0.0 pp); Codex leaves catchable errors (OO +12.9 pp).
- Gains track the reviewer's strength relative to the writer, consistent with prior self-repair results; a second pass can subtract as well as add.
- Most AO losses are rewrites of working drafts (an interpretive pattern from inspected cases, not a coded taxonomy). A reviewer prompt that first decides whether to intervene, and keeps the writer's structure by default, is a concrete next step.
- A reviewer able to run even public tests could recover much of what static inspection misses; other model pairs are untested.

9. Cost and latency trade-offs

Pass rate versus cost per task



Pass rate versus latency per task



Circles: solo baselines; diamonds: with reviewer; arrows: the reviewer's effect (green gain, red loss, gray flat); single-run per-task means at current pricing. No reviewed condition beats Claude Opus 4.7 solo on both axes. OA buys each net fixed task for about $1.40; OO about $0.95.

10. Practical guidance

On this benchmark, with one static review pass:

- **Writer must be Codex GPT-5.5:** use Claude Opus 4.7 review (OA) when accuracy dominates (89.7%; the lowest reviewed regression rate, 4.3%). Use self-review (OO) when average per-task latency must stay near 70 s (84.5%).
- **Writer must be Claude Opus 4.7:** skip review. Self-review left the pass rate unchanged in this run (3 fixes, 3 regressions) while adding 72% to cost and 49.6 s of latency; Codex GPT-5.5 review loses 8.6 pp.
- **Both ends free:** Claude Opus 4.7 solo is Pareto-optimal; AA is strictly dominated by A.

Scope and limitations

116 tasks; a single run per condition; one benchmark family; one static review pass with no execution feedback; results are sensitive to prompts and to model and tool versions; reproducing all six conditions costs about $225.

A second review pass is not free insurance: its value depends on who wrote the draft and who reviews it.

Code, data, prompts, and per-task records: **github.com/shawnzxiang/cross-model-review-code** · Supported by US NSF Award #2436427